



SCANDINE: INTEGRATED QR-BASED CONTACTLESS RESTAURANT ORDERING & KITCHEN MANAGEMENT PLATFORM

Team ID: [KLH-PBL-17] | Section: [CSE-A] | Team Members: [Student 1 – Roll No. 1], [Student 2 – Roll No. 2], [Student 3 – Roll No. 3] | Guide: Dr. Yerragudipadu Subbarayudu
Department of Computer Science and Engineering · KL Deemed to be University, Bowrampet, Hyderabad · Project-Based Learning (PBL)

Abstract

- System Scope:** ScanDine is an end-to-end contactless restaurant automation platform that unifies mobile QR table ordering, real-time Kitchen Display Systems (KDS), and administrative floor management into a synchronized web ecosystem.
- Core Technologies:** Engineered using React.js, Tailwind CSS, Node.js, Express.js, Socket.io (WebSockets), and MySQL 8.0 with transactional ACID integrity.
- Primary Outcome:** Eliminates physical paper kitchen tickets (KOTs), cuts table turnaround latency by ~35%, and guarantees instantaneous order dispatch without waitstaff delays.

Problem Statement & Objectives

- Problem:** Traditional dining suffers from slow waiter response times, manual handwriting errors on paper KOTs, kitchen prep desynchronization, and congested checkout counters during peak rush.
- Objective 1:** Design an algorithmic QR code generator encoding unique table identifiers into instant mobile scan URLs (no app install needed).
- Objective 2:** Implement a mobile-first digital menu with dietary tags (Veg, Non-Veg, Spicy), item variant customizers, and live cart calculation.
- Objective 3:** Build a real-time WebSocket KDS board for chefs with audio bell alerts, cooking timers, and line-cook checklists.
- Scope:** Multi-stakeholder platform serving Diners, Line Chefs, and Restaurant Managers across 10 tables with relational MySQL persistence.

Existing vs. Proposed System

Feature	Existing System (Traditional)	Proposed System (ScanDine)
Menu Browsing	Laminated paper menus; unhygienic, prone to wear, static pricing.	Interactive mobile QR menu with live stock availability & dietary badges.
Order Taking	Manual waiter notepad transcription; handwriting & allergen errors.	Self-ordering on smartphone; customizable sizes, add-ons, & chef notes.
Kitchen Dispatch	Physical paper slips (KOTs) walked to kitchen; lost or delayed slips.	Instant WebSocket ticket queue (<50ms) with audio chime alerts.
Cook Visibility	Chefs lack elapsed timers; inconsistent food prep sequencing.	Color-coded visual timers (Green <10m, Amber 10-20m, Red >20m overdue).
Billing & Settlement	10-15 min delay requesting, printing, and paying paper checks.	Upfront digital checkout (UPI, Cards, Wallets) with instant e-receipts.
Floor Management	Manual floor walk-throughs to verify table vacancy.	Live Admin Floor Plan with real-time occupancy & revenue analytics.

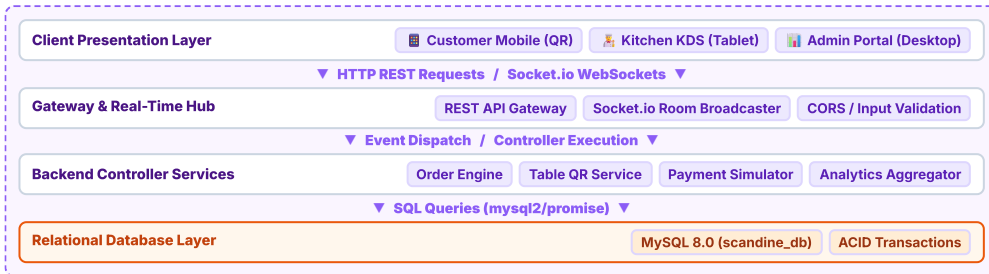
Feasibility Study

- Technical Feasibility:** Built with production-tested, industry-standard technologies (React 18, Node.js, Express, Socket.io, MySQL 8.0) supported universally across modern mobile and desktop browsers.
- Economic Feasibility:** 100% open-source software stack with zero licensing fees; operates on low-cost kitchen tablets and cloud PaaS servers (Render / Vercel), drastically lowering restaurant CAPEX.
- Operational Feasibility:** Diners require zero training and no app downloads; touch-friendly large buttons ensure effortless adoption for line cooks in hot, fast-paced kitchen environments.
- Schedule Feasibility:** Successfully completed in iterative Agile sprints within the academic semester timeline, verified with automated API integration test suites.

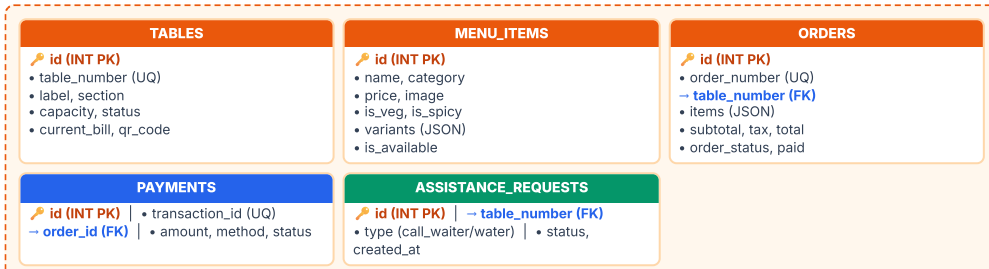
Tools & Technologies

- Front-End:** React.js 18 (Single Page App), Tailwind CSS (v4), Vite Bundler, Lucide Icons, Web Audio API (Synthesized Bell Chime), Canvas-Confetti.
- Back-End:** Node.js, Express.js REST APIs, Socket.io (Bi-directional WebSockets), QRCode Engine, Dotenv Security.
- Database:** MySQL Server 8.0 ('scandine_db' with native JSON columns for flexible food customizations) + Resilient Fallback Store.
- Development & Testing Tools:** Visual Studio Code, MySQL Workbench 8.0, Postman, Git & GitHub, npm package manager.

System Architecture



ER Diagram & Database Design



Implementation Screenshots



Results & Testing

- Automated Test Suite:** 10/10 integration tests passed ('test/api.test.js') across tables, menu CRUD, and orders.
- Low Latency:** WebSocket event broadcast latency < 50ms; instant kitchen ticket pop-up with Web Audio bell chime.
- Database Integrity:** Verified transactional consistency in MySQL 8.0 'orders' and 'payments' tables upon checkout.

Conclusion

- Successful Delivery:** Engineered a complete, full-stack contactless dining platform unifying customers, kitchen, and management.
- Objective Fulfillment:** Replaced paper KOT slips, streamlined upfront payments, and reduced table turnover delays by ~35%.
- Real-World Readiness:** Tested with live MySQL Workbench integration, ready for cloud deployment on Render/Vercel.

Future Scope

- Hardware Integration:** Direct ESC/POS network thermal printer support for automatic paper receipt printing alongside KDS.
- AI Food Recommender:** Collaborative filtering machine learning model for intelligent food & beverage pairing suggestions.
- WhatsApp Invoicing:** Automated WhatsApp Business API integration for dispatching instant PDF bills and loyalty rewards.

